Sri Lanka Institute of Information Technology



Lab Submission
<Lab sheet 04>

<IT26100802>

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Discrete Mathematics | IT1160

B.Sc. (Hons) in Information Technology

Part 01)

01.

In [5]:
```
for n in range(1,11):
    a = n**2+3*n

    print(a)
```

```
4
10
18
28
40
54
70
88
108
130
```

In [6]:
```
total = 0

for n in range(1,11):
    a = n**2+3*n
    total += a

print(total)
```

```
550
```

02.

In [7]:
```
bacteria = 300

for hour in range(0, 11):
    print(bacteria)
    bacteria = bacteria * 2
```

```
300
600
1200
2400
4800
9600
19200
38400
76800
153600
307200
```

In [8]:
```
bacteria = 300

for hour in range(10):
    bacteria = bacteria * 2

print(bacteria)
```

```
307200
```

03.

In [9]:
```
storage = 5

for month in range(1, 16):
    print(storage)
    storage = storage + 4
```

```
5
9
13
17
21
25
29
33
37
41
45
49
53
57
61
```

In [10]:
```
storage = 5
total = 0

for month in range(1, 16):
    total = total + storage
    storage = storage + 4

print(total)
```

```
495
```

iii) 495GB

04.

```
495
```

In [11]:
```
total = sum((-2)**j for j in range(0, 5))
print(total)
```

```
11
```

In [12]:
```
total = sum(3 for i in range(1, 11))
print(total)
```

```
30
```

In [14]:
```
total = sum(2**(j+1) - 2**j for j in range(0, 9))
print(total)
```

```
511
```

05.

```
511
```

In [15]:
```
import math

print(math.factorial(5))
```

```
120
```

Part 02)

01.

In [16]:
```
a = 5

for i in range(15):
    print(a)
    a = a + 4
```

```
5
9
13
17
21
25
29
33
37
41
45
49
53
57
61
```

In [17]:
```
a = 5
total = 0

for i in range(15):
    total = total + a
    a = a + 4

print(total)
```

```
495
```

02.

In [18]:
```
salary = 720000

for year in range(1, 11):
    print(salary)
    salary = salary + 45000
```

```
720000
765000
810000
855000
900000
945000
990000
1035000
1080000
1125000
```

In [19]:
```
salary = 720000
total = 0

for year in range(1, 11):
    total = total + salary
    salary = salary + 45000

print(total)
```

```
9225000
```

03.

In [20]:
```
population = 100

for year in range(1, 11):
    print("Year", year, ":", population)
    population = population * 1.15 - 5
```

```
Year 1 : 100
Year 2 : 109.99999999999999
Year 3 : 121.49999999999997
Year 4 : 134.72499999999997
Year 5 : 149.93374999999995
Year 6 : 167.4238124999999
Year 7 : 187.53738437499987
Year 8 : 210.66799203124984
Year 9 : 237.2681908359373
Year 10 : 267.85841946132786
```

In [21]:
```
population = 100
total_increase = 0

for year in range(2, 11):
    new_population = population * 1.15 - 5
    increase = new_population - population
    total_increase = total_increase + increase
    population = new_population

print(total_increase)
```

```
167.85841946132786
```

Year 9 = 237.268...

Year 10 = 267.858...

04.

```
In [23]: total = sum(2*r + 1 for r in range(50, 100))
         print(total)
```

```
7500
```

```
In [24]: total = sum(4 * 3**r for r in range(1, 6))
         print(total)
```

```
1452
```

```
In [25]: total = sum((-1)**n * n**2 for n in range(1, 5))
         print(total)
```

```
10
```

05.

```
In [27]: total = 0

         for i in range(1, 6):
             packets = 3*i + 2
             total = total + packets

         print(total)
```

```
55
```

55